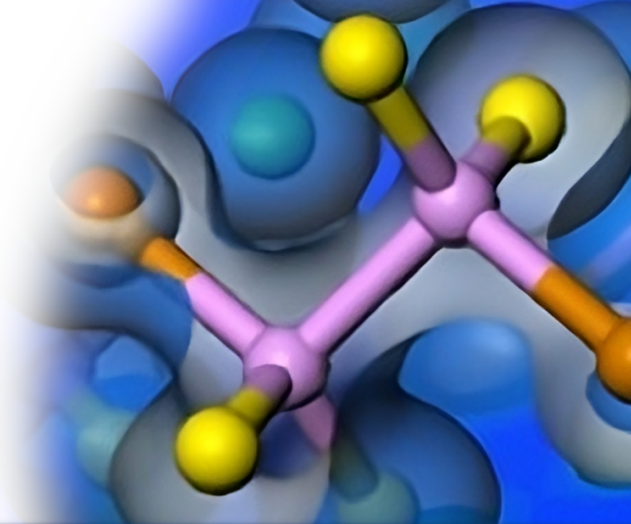




Electronic structure, plane waves
and pseudopotentials

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Introduction

DFT

Bloch's Theorem

Reciprocal-Space
Sampling

Plane-waves

Pseudopotentials

Summary

- Aim: predict electrons' behaviour from *first principles*
i.e. no knowledge of what they'll do beforehand
- Can use quantum mechanics
- Time-dependent Schrödinger equation is difficult!
- Many-body wavefunction $\Psi = \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N; t)$



The Born-Oppenheimer Approximation ♣ CASTEP

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Compared to electrons, nuclei are massive and slow:

- Electrons react 'instantly' to nuclear motion
- Nuclear wavefunctions are extremely localised



An Ode to Born-Oppenheimer



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*Electrons are light,
they travel at pace,
with wavefunctions spreading all over the place;*

*Nuclei are not,
they're massive and slow,
you don't need Schrödinger to see where they go.*



The Born-Oppenheimer Approximation ♣CASTEP

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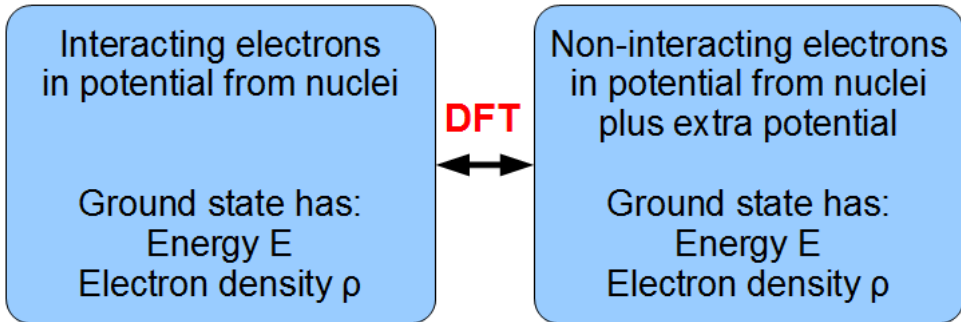
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Summary

- Only use quantum mechanics for electrons
- Only use time-*independent* Schrödinger equation.
- Still difficult!

In real life the behaviour is dominated by the ground state. Can this help us?





In the Kohn-Sham scheme, we write

$$E_{kin} + E_{e-e} = E_{kin}^{non-int.} + E_{e-e}^{non-int.}[\rho] + E_{xc}[\rho]$$

The term $E_{xc}[\rho]$ is the **exchange-correlation functional**.

Can compute $\rho(\mathbf{r})$ from wavefunctions ψ_b ,

$$\rho(\mathbf{r}) = \sum_b |\psi_b(\mathbf{r})|^2$$



We now solve the Schrödinger equation

$$\hat{H}[\rho] \psi_b(\mathbf{r}) = \epsilon_b \psi_b(\mathbf{r}),$$

where

$$\hat{H}[\rho] = -\frac{\hbar^2}{2m} \nabla^2 + \hat{V}_{e-e}^{non-int.}[\rho] + \hat{V}_{e-n}[\rho] + \hat{V}_{xc}[\rho].$$

and particle b goes in the b th solution.



We now solve the Schrödinger equation

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and particle b goes in the b th solution. Unfortunately we don't know what $\hat{V}_{xc}[\rho]$ is!



- Only $E_{xc}[\rho]$ unknown $\sim 10\%$ of E
- If electrons evenly spread in space *can* compute $E_{xc}[\rho]$
- Use this $E_{xc}[\rho]$ as approx. to true E_{xc} in our calculations
- Contribution to E_{xc} from $\mathbf{r}$ only depends on $\rho(\mathbf{r})$
- This is the *Local Density Approximation (LDA)*

$$E_{xc}[\rho](\mathbf{r}) \approx E_{xc}^{\text{LDA}}(\rho(\mathbf{r}))$$



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- More advanced approximations exist
- Many use the *gradient* of ρ too
- Called *Generalized Gradient Approximations (GGAs)*
- One of best from Perdew, Burke and Ernzerhof (PBE)
- Some functionals also include the kinetic energy density (related to the curvature of the density)
- Called *meta-GGAs*
- A good recent method is RSCAN



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Started with a time-dependent, many-nuclei, many-electron quantum mechanics problem.

Ended with static, single-particle quantum mechanics:

$$\hat{H}[\rho]\psi_b = \epsilon_b\psi_b$$

Just need to calculate enough states for our electrons.



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Summary

- Even 'perfect' DFT is only exact for the ground state
- Exchange-correlation is approximated crudely
- No collective electron-nuclear motion
- No nuclear quantum effects (e.g. zero-point motion)



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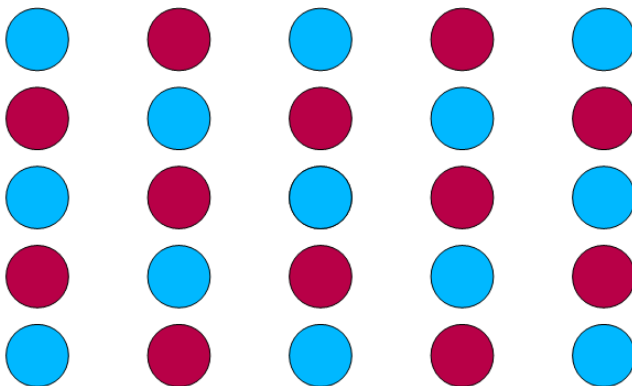
Summary

- Just need to solve a single-particle equation
- Need one state per electron
- Even a few grams of material has over 10^{23} electrons
- We need a *lot* of states!

This could take a while...



In the solid state, most materials like to have their atoms arranged in some kind of regular, repeating pattern, e.g.



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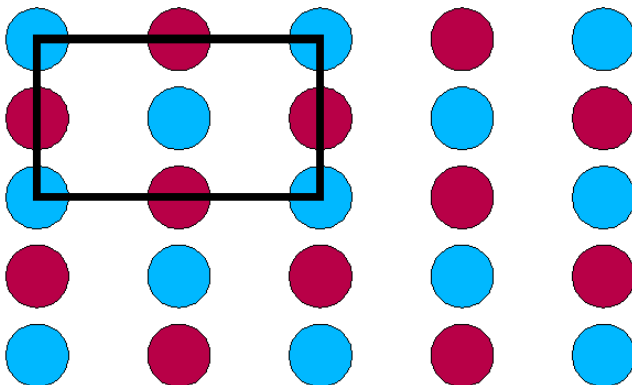
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In the solid state, most materials like to have their atoms arranged in some kind of regular, repeating pattern, e.g.



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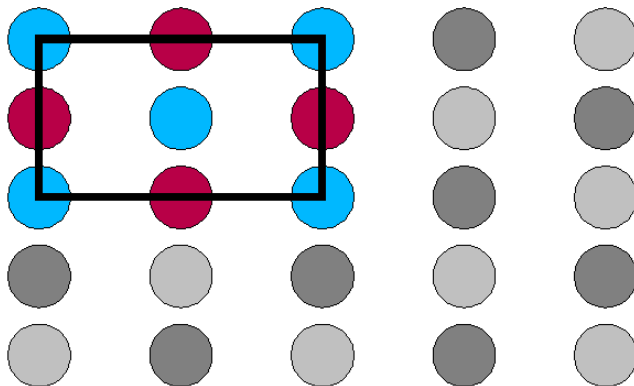
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In the solid state, most materials like to have their atoms arranged in some kind of regular, repeating pattern, e.g.



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If the nuclei are arranged in a periodically repeating pattern, their potential acting on the electrons must also be periodic.

$$V(\mathbf{r} + \mathbf{L}) = V(\mathbf{r})$$

where $\mathbf{L}$ is any lattice vector.

What does this mean for the density and wavefunction?



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Summary

If the potential is periodic, then so is the density:

$$\rho(\mathbf{r} + \mathbf{L}) = \rho(\mathbf{r})$$

What about the wavefunction?



If the potential is periodic, then so is the density:

$$\rho(\mathbf{r} + \mathbf{L}) = \rho(\mathbf{r})$$

What about the wavefunction?

$$\rho(\mathbf{r}) = |\psi(\mathbf{r})|^2$$

so if $\rho(\mathbf{r})$ is periodic, then so is the magnitude of the wavefunction.

Remember wavefunctions are complex; their magnitude is periodic, but their phase can be anything.



Bloch's theorem: in a periodic potential, the density has the same periodicity. The possible wavefunctions are all 'quasi-periodic':

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_k(\mathbf{r}),$$

where $u_k(\mathbf{r} + \mathbf{L}) = u_k(\mathbf{r})$, and $e^{i\mathbf{k}\cdot\mathbf{r}}$ is an arbitrary phase factor.

$$\begin{aligned}\psi_k(\mathbf{r} + \mathbf{L}) &= e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{L})} u_k(\mathbf{r} + \mathbf{L}) \\ &= e^{i\mathbf{k}\cdot\mathbf{L}} \psi_k(\mathbf{r})\end{aligned}$$



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In principle, have to integrate over all $\mathbf{k}$
BUT wavefunctions change slowly with $\mathbf{k}$



In principle, have to integrate over all **k**
BUT wavefunctions change slowly with **k**
→ can approximate integral with a sum:

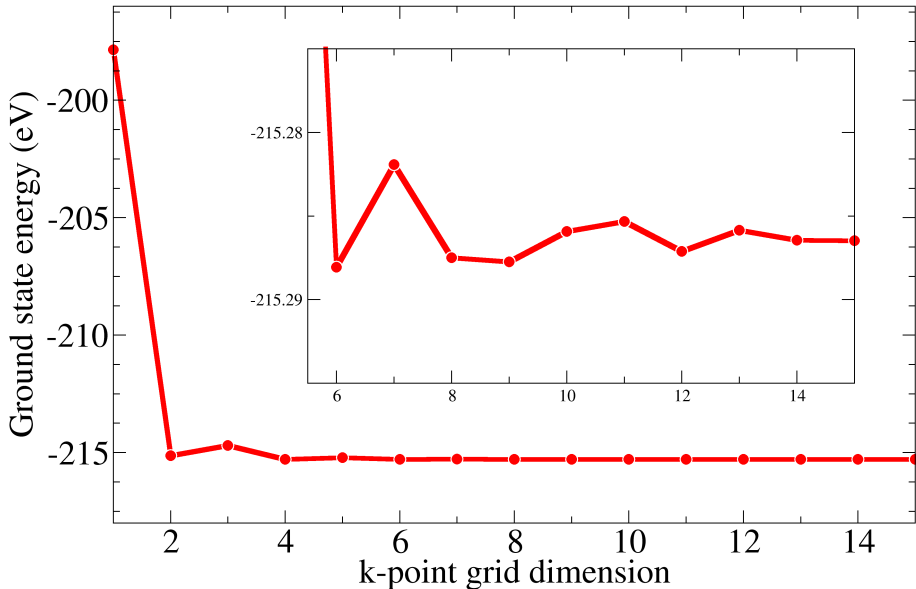
$$\begin{aligned}\rho(\mathbf{r}) &= \int |\psi_{\mathbf{k}}(\mathbf{r})|^2 d^3\mathbf{k} \\ &\approx \sum_{\mathbf{k}} |\psi_{\mathbf{k}}(\mathbf{r})|^2\end{aligned}$$

Need to ensure enough **k**-points to get accurate results.

We'll be looking at **k**-points more closely in a later talk.



k-point sampling





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We need to choose a suitable basis set to represent our wavefunctions, but what should we choose...

- Points on a grid?
- Polynomials?
- Gaussians?
- Atomic orbitals?

None of these reflect the periodicity of our problem.



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$u_k(\mathbf{r})$ is periodic $\longrightarrow$ express as 3D Fourier series:

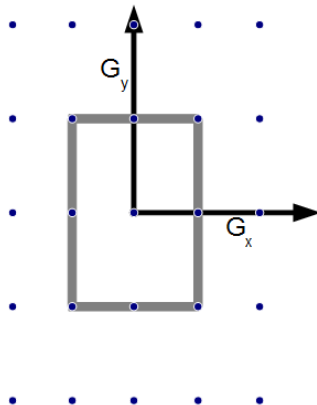
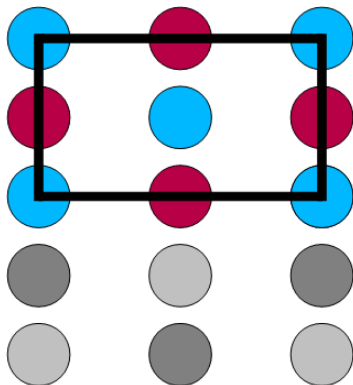
$$u_k(\mathbf{r}) = \sum_{\mathbf{G}} c_{Gk} e^{i\mathbf{G} \cdot \mathbf{r}}$$

Only **reciprocal lattice vectors** have the right periodicity: $\mathbf{G} \cdot \mathbf{L} = 2\pi m$
(where m is an integer).

If we make the cell longer in one direction, the allowed wavevectors in that direction become shorter.



Wavevectors have units of $1/\text{length}$ $\rightarrow$ describe points in **reciprocal-space**. We can plot the allowed wavevectors in this space:





So we have:

$$u_k(\mathbf{r}) = \sum_{\mathbf{G}} c_{Gk} e^{i\mathbf{G}\cdot\mathbf{r}}$$

$e^{i\mathbf{G}\cdot\mathbf{r}}$ is plane-wave travelling perpendicular to $\mathbf{G}$.

Infinite $\mathbf{G}$, but c_{Gk} become smaller as $|\mathbf{G}|^2$ becomes larger.

Define a cut-off energy $E^{cut} = \frac{\hbar^2}{2m} |\mathbf{G}|^2$ and only include plane-waves with energies less than this cut-off.



Cut-off Energy

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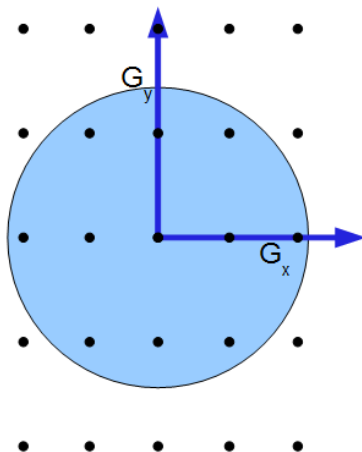
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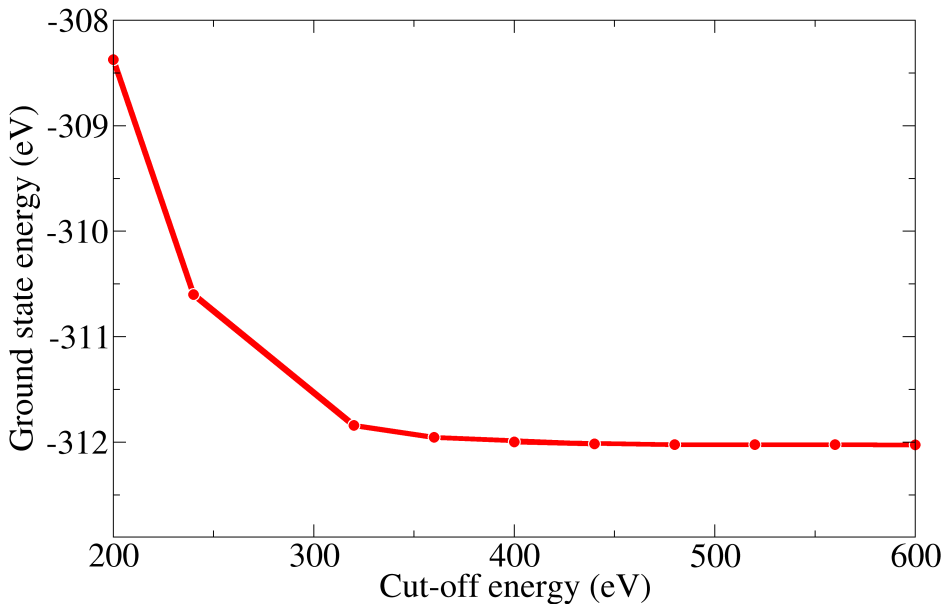
Summary

- Must ensure E^{cut} high enough to give accurate results
- Recalculate with higher E^{cut} until properties converge

How do the calculated results depend on the cut-off energy?



Cut-off Energy





Cut-off Energy

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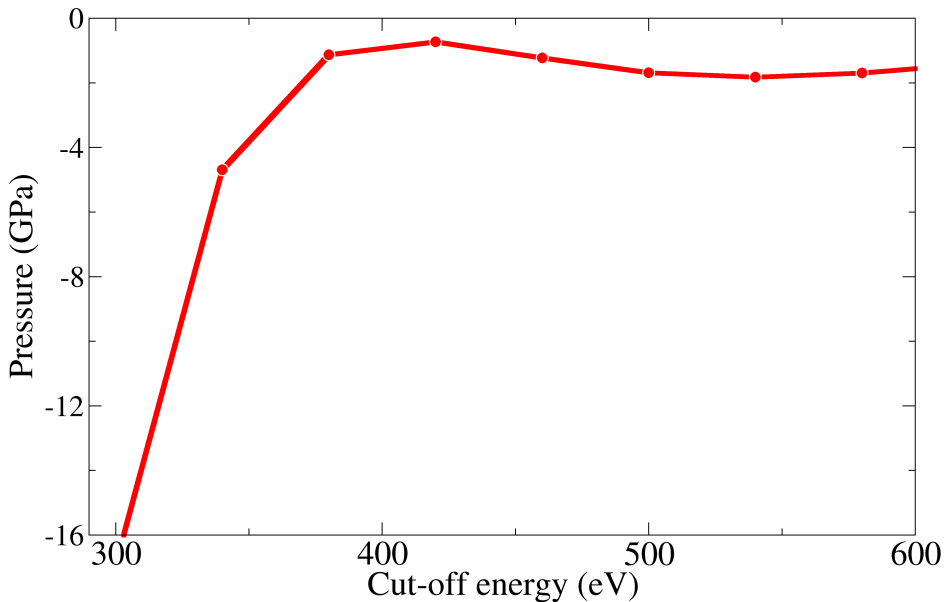
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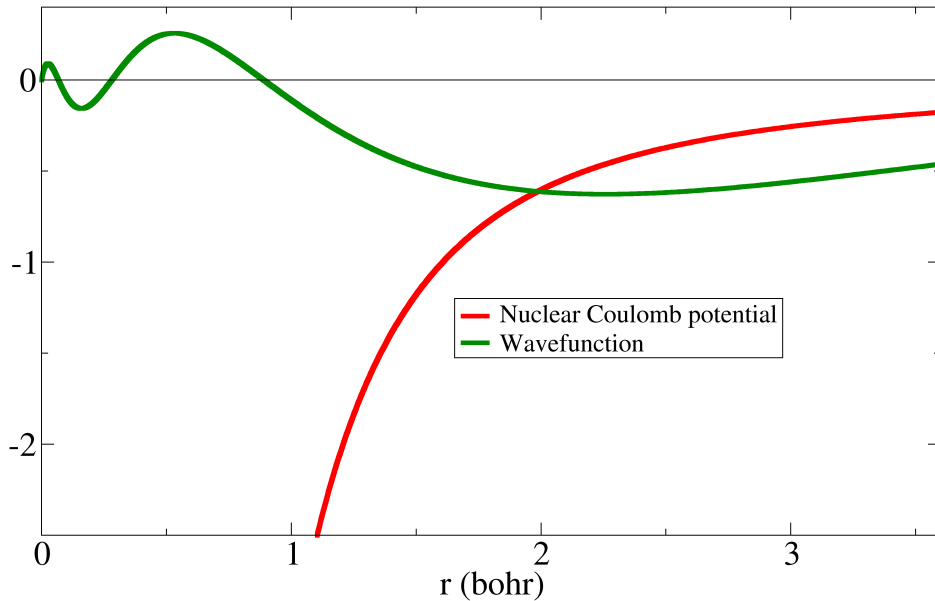
Summary

The very large **G** components describe the region of space where the wavefunction is varying very quickly.

This happens when the potential is very attractive – the strongest potentials are those near the nuclei.



Pseudopotentials





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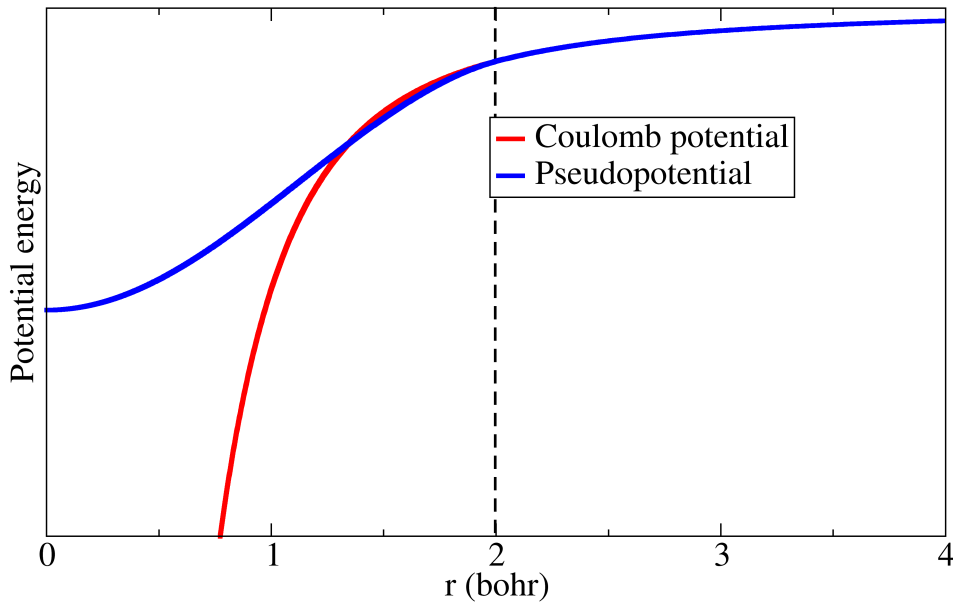
$\psi(\mathbf{r})$ near nuclei not very interesting
– don't affect the chemical properties very much.

Can replace the Coulomb potential with a weaker potential. This modified potential is called a **pseudopotential**.

ψ doesn't vary as quickly $\rightarrow$ can use smaller E^{cut} .



Pseudopotentials





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Summary

- Core electrons entirely near nuclei
- Repel outer electrons, otherwise don't affect the chemical properties etc.
- Provided we reproduce this screening effect, we don't need to compute the core electrons!
- nucleus + core electrons = ion
- Make pseudopotential with same effect on the outer electrons



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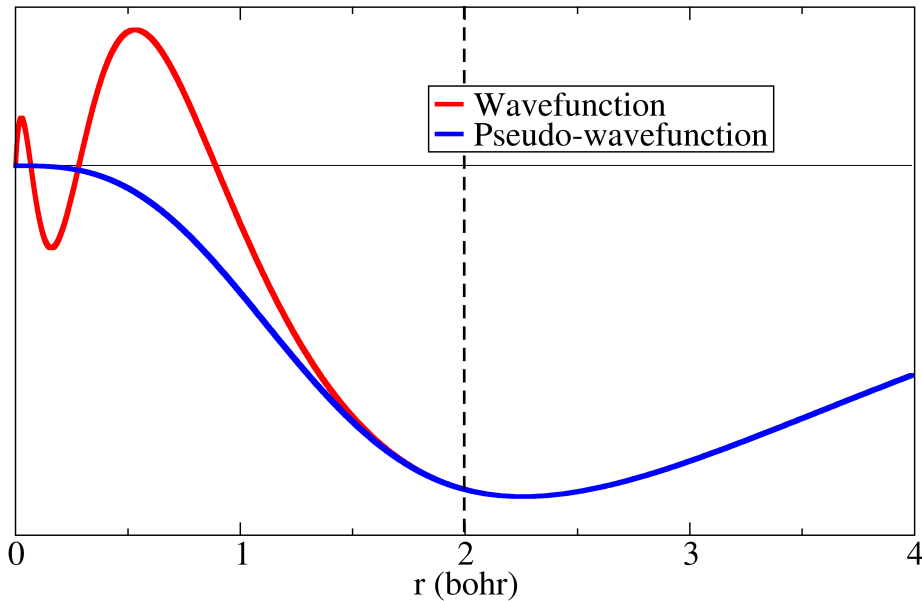
Pseudopots

Summary

- Core electrons entirely near nuclei
- Repel outer electrons, otherwise don't affect the chemical properties etc.
- Provided we reproduce this screening effect, we don't need to compute the core electrons!
- nucleus + core electrons = ion
- Make pseudopotential with same effect on the outer electrons
- Have reduced E^{cut}
- Have removed explicit treatment of core electrons
→ Fewer states needed from Schrödinger equation.

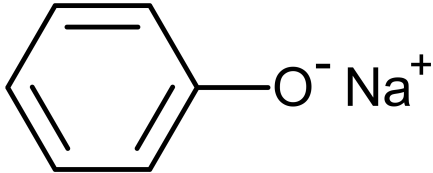


Pseudopotentials





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- Summary





CASTEP uses a plane-wave basis for the wavefunctions, and these are sampled at discrete $\mathbf{k}$ -points. You need to converge your answers with respect to:

- Cut-off energy for plane-waves
- $\mathbf{k}$ -point sampling

Pseudopotentials accelerate the cut-off energy convergence and reduce the number of electrons we have to consider.

The ground state is found by improving an initial guess iteratively until the change in energy is small.